F7AB - Actuarial and Financial Mathematics

Tutorial 8 – Fixed Interest Bonds, Tax and Index-Linked Securities

Question 1 A government has just issued a fixed interest security (or bond) of nominal amount £100 pays a coupon of 6% p.a. at the end of each of the next ten years and is redeemed at par at the end of the ten years.

1. Determine the price of the security which gives a redemption yield of 6% p.a. effective to a purchaser who pays no tax. Explain the answer.

2. Calculate the price of the security that gives a redemption yield to the same purchaser of 8% p.a. effective.

Question 2 Suppose that in the above example the coupon is £6 p.a. payable half-yearly.

1. Calculate the price of the security that gives a yield to the same purchaser of: (a) 6% p.a. effective, and (b) 8% p.a. effective.

2. Explain why your result in (a) is not £100.

Question 3 A fixed interest security of nominal amount £100 000 will be repaid at 110% after 15 years. The security bears a coupon of 9% p.a. payable annually in arrears. On the issue date, the security is purchased for £80 000 by an investor who is liable to income tax at 40% and to capital gains tax at 30%.

1. Calculate the investor's net (meaning "after tax") effective redemption yield p.a. for the transaction.

2. The investor wishes to secure an after-tax redemption yield of 9% p.a. Determine how much the investor should pay for the security.

Question 4 Consider a 20-year bond, bearing a half-yearly coupon of 8% per annum and redeemable at 120%. It is exactly six months until the next coupon. Calculate the price of the bond to give a redemption yield of 9% per annum effective after tax to an investor who pays:

1. no taxes.

2. income tax at 20% but no capital gains tax (CGT).

3. income tax at 20% and CGT at 30%.

Question 5 On 1 January 2003, a corporation issues a number of bonds, each of nominal amount £100, to be redeemed at 98% on 1 January 2028. The coupon is 7% per annum, payable half-yearly on each 1 January and 1 July.

1. The bond is priced at issue to give a redemption yield of 8.16% per annum to a non-tax-paying investor. Calculate the issue price per £100 nominal.

2. Anna, who pays income tax at 25% and CGT at 40%, bought some of the bonds on the issue date. On 1 July 2013, just after the coupon then due was paid, she sold the bonds to Bob, who pays both income tax and CGT at 40%. Bob paid a price that gave him a redemption yield after taxes of 6% p.a. effective. Calculate the price that Bob paid per £100 nominal.

3. Calculate the yield realised by Anna on her completed transaction.

Question 6 Investors often allow for risk by requiring a higher yield to lend money to a borrower who is thought to be at higher risk of not repaying some or all of the interest and capital. Explain the consequences for companies or governments who are trying to raise money in the bond markets.

Question 7 An investor buys a 20-year bond, to be redeemed at 110% and bearing semi-annual coupons at rate 6% per annum, to obtain a redemption yield of 5% p.a. effective. The investor pays income tax at 25% but no CGT.

1. Calculate the price paid by the investor.

2. Just after the 20th coupon payment, the investor's rate of income tax drops to 20%. If she holds the bond to redemption, calculate her realised yield on the whole transaction.

Question 8 A fixed interest stock with nominal value of £100 was issued on 1 January 1995. The coupon rate was 4% p.a. payable half-yearly in arrears for the first 10 years and then 3% p.a. payable half-yearly in arrears for the next 10 years. Redemption was at par after 20 years. On 1 July 1999, just after the July dividend was paid, the stock was quoted at a price which would have given a tax-free investor a redemption yield of 6% p.a. The stock was bought on that date by an investor who was subject to income tax at 40% and to capital gains tax at 20%.

1. The investor held the bond until the redemption date. Calculate his after tax redemption yield.

2. On 1 August 2000 the stock was priced to give an after tax redemption yield of 5% p.a. effective to the taxed investor described above. Calculate the price of the stock on 1 August 2000. (Note: be careful with the dates.)

Question 9 State the characteristics of index-linked government bonds.

Question 10 On 9 October 2007 an investor, who is not liable to tax, had the choice of purchasing either:

(A) $7\frac{1}{2}\%$ Treasury Stock at price £107 per £100 nominal repayable at par.

(B) 2% Index-Linked Treasury Stock 2016 at a price of £207 per £100 nominal.

The RPI base figure for indexing is 69.5 and the next coupon is to be paid on 9 April 2008. The time lag for inflation indexing is 8 months.

Both stocks pay interest half-yearly in arrears and are redeemed on 9 October 2016. The latest published value of the RPI is 207.3, which is the RPI for August 2007.

Assuming that today is 9 October 2007 and the RPI will grow continuously at a rate of 2.5% per annum from its latest published value,

1. Verify that the real yield from stock (A) as at 9 October 2007 is 4% per annum effective.

2. By considering the series of future receipts from stock (B), derive a formula for the present real value of £100 nominal of that stock.

3. Calculate the real yield as at 9 October 2007 from stock (B) by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Question 11 On 15 March 2006 the government of a country issued an index-linked bond of term 6 years. Coupons are payable half-yearly in arrears, and the annual nominal coupon rate is 3%.

Interest and capital payments are indexed by reference to the value of an inflation index with a time lag of 8 months.

A tax-exempt investor purchased the stock at £111 per £100 nominal on 16 September 2009, just after the coupon payment had been made.

You are given the following values of the inflation index:

Date	Jul 2005	Mar 2006	Jul 2009	Sep 2009
Inflation Index (Date)	110.5	112.1	126.7	127.4

1. Calculate the amount of the coupon payment per £100 nominal stock on 15 March 2010.

2. Calculate the effective real annual yield to the investor on 16 September 2009. You should assume that the inflation index will increase at a constant rate of 4% per annum effective from its value in September 2009.

3. Without doing any further calculations, explain how the effective real annual yield would alter, if at all, if the inflation index for July 2005 had been more than 110.5 and all the other index values were unchanged.

Answers: Q1.1. £100; Q2.1. £86.57984; Q2.1(a). £100.65278; Q2.1(b). £87.37017; Q3.1. 0.0772; Q3.2. £70.47126; Q4.1. £96.04715; Q4.2. £81.12001; Q4.3. £78.92111; Q5.1. £88.97748; Q5.2. £79.42754; Q5.3. 5.12% p.a.; Q7.1. £98.23044; Q7.2. 5.001% p.a.; Q8.1. 3.999% p.a.; Q8.2. £67.17545; Q10.3. 6.481%; Q11.1. £1.71991; Q11.2. 3.52% p.a.